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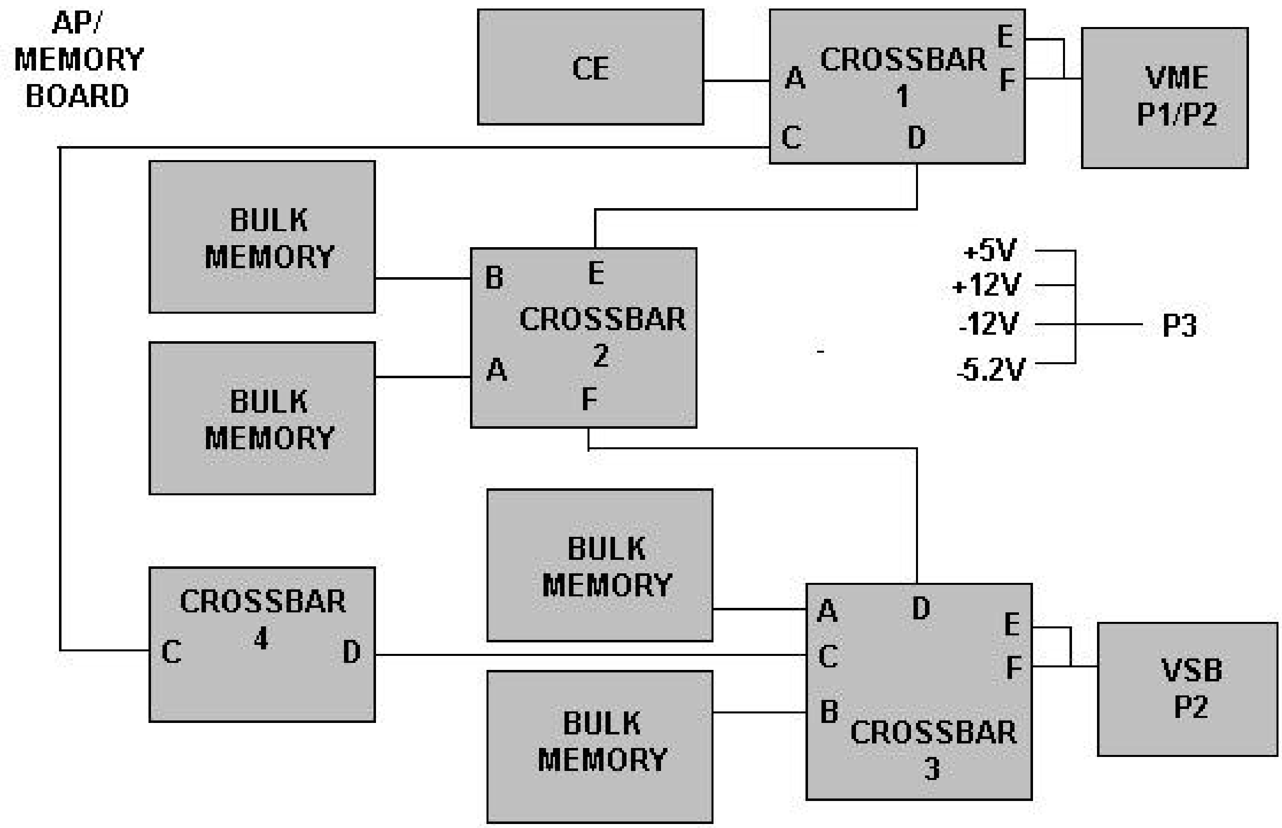
1 - AP CE

Refer to Illustration 1-1 for the AP/Memory Board schematic.



AP/MEMORY BOARD SCHEMATIC
ILLUSTRATION 1

The AP/Memory Board uses standard RACE building blocks wherever possible (this includes RACE CE modules, Crossbar ASICs, and the VME interface). The AP/Memory Board mothercard is a VME compatible 9U x 400 mm printed circuit card.

The architecture of the basic AP/Memory Board consists of multiple crossbar ASICs interconnecting a VME bus interface, a VSB bus interface, connectors for up to three standard, two processor (or I/O) daughtercards (supported by six crossbar connections), and connectors for up to four bulk memory (128 Mbyte) daughtercards. For low end systems, standard 32-Mbyte memory daughtercards (large memory CEs with the i860 removed) are used as the bulk memory card. Due to differences in form factor, a 32-Mbyte memory card occupies two bulk memory locations (but uses only one crossbar port).

2 - AP/MEMORY CROSSBAR

Each crossbar supports three concurrent transfer paths. Any transfer in the system may take place concurrently with any other as long as the two transfers do not use the same port on any crossbar in the path. Conflicts are resolved by the hardware.

Transfers requested by the VSB slave interface have priority through the crossbars. A crossbar transfer generated by VSB master accessing the bulk memory causes any other crossbar transfers to suspend until the VSB/crossbar access is complete. Since the VSB interface contains a FIFO, crossbar transfers use the bandwidth efficiently. Because of their lower priority, VME transfers to /from the bulk memory can be suspended for extended periods by VSB accesses. For this reason, the VME bus controller should have a bus timeout value greater than 20 milliseconds to prevent access conflicts from causing VME bus timeouts.

3 - AP/MEMORY VME BUS

The VME interface is a full-functioned master/slave interface.

4 - BULK MEMORY

The Bulk Memory cards are DRAM daughtercards with characteristics as described below. The Bulk Memory card interface is a standard crossbar port.

128 Mbyte Bulk Memory Features

- 128 Mbytes of memory.
- 32-bit parity (1 parity bit for 4 bytes of data).
- DMA controller programmable from any crossbar master.

32 Mbyte Bulk Memory Features

- 32 Mbytes of memory.
- 4 parity bits for 4 bytes of data (same memory requirements as byte parity).
- DMA controller programmable from any crossbar master.

5 - AP/MEMORY VSB BUS

The VSB interface is a master/slave interface designed to a subset of the VSB specification. Only those features required for high performance data transfers are supported by the interface.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 3, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.